

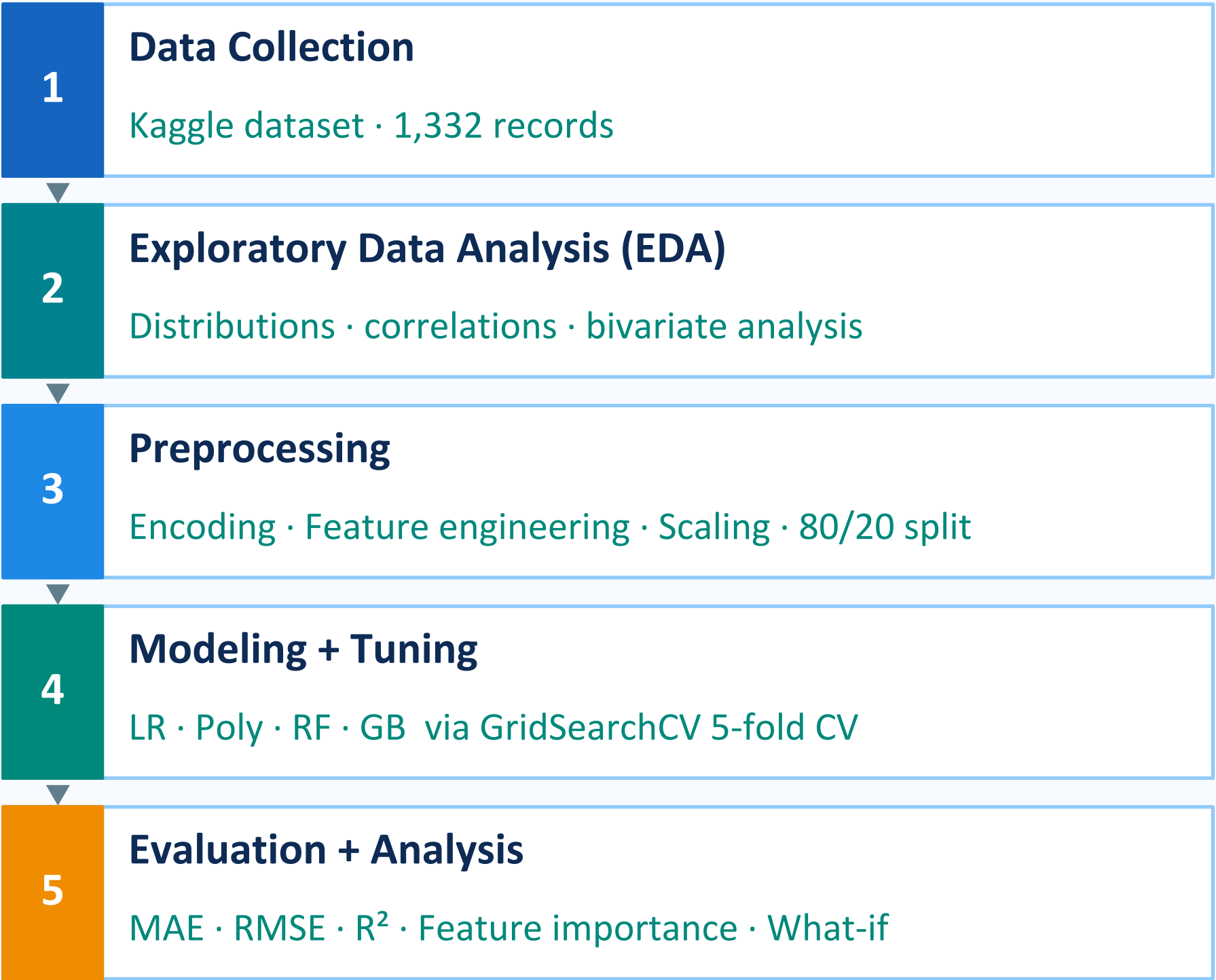
Using Machine Learning to Predict Healthcare Insurance Cost

Dataset & Research Pipeline

Data Variables

Variable	Type	Values
age	Numerical (continuous)	18–60
gender	Categorical (binary)	Male / Female
bmi	Numerical (continuous)	16.0–53.1
bloodpressure	Numerical (continuous)	80–140
diabetic	Categorical (binary)	Yes / No
children	Numerical (discrete)	0–5
smoker	Categorical (binary)	Yes / No
region	Categorical (nominal)	NE / NW / SE / SW
claim ★	Numerical (continuous)	\$1,121–\$63,770 TARGET

Research Pipeline



EDA — Variable Distributions

Age: 38.09 yrs

mean · range 18–60

BMI: 30.66

mean · obese threshold ≥ 30

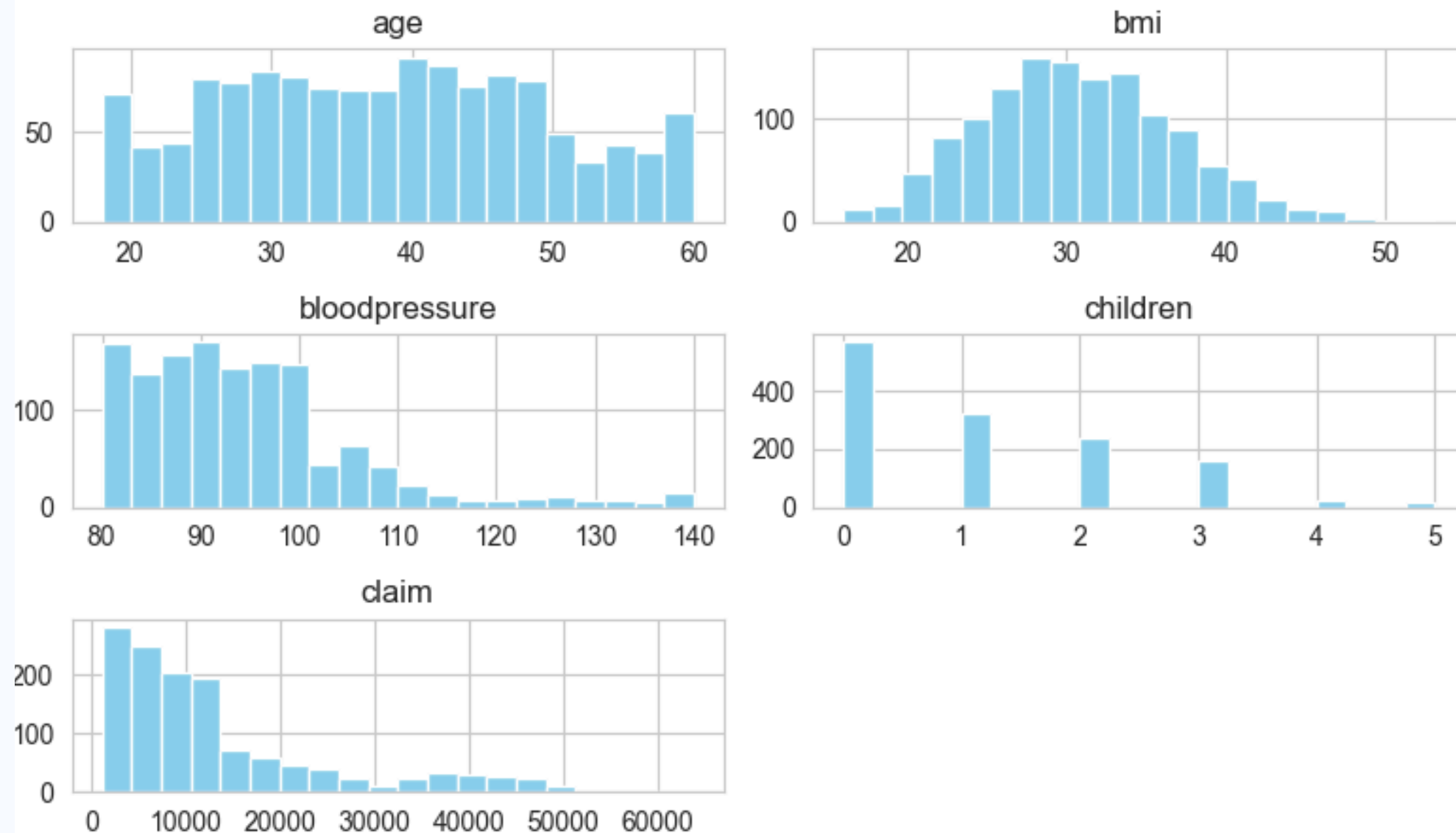
BP: 94.19 mmHg

mean · range 80–140

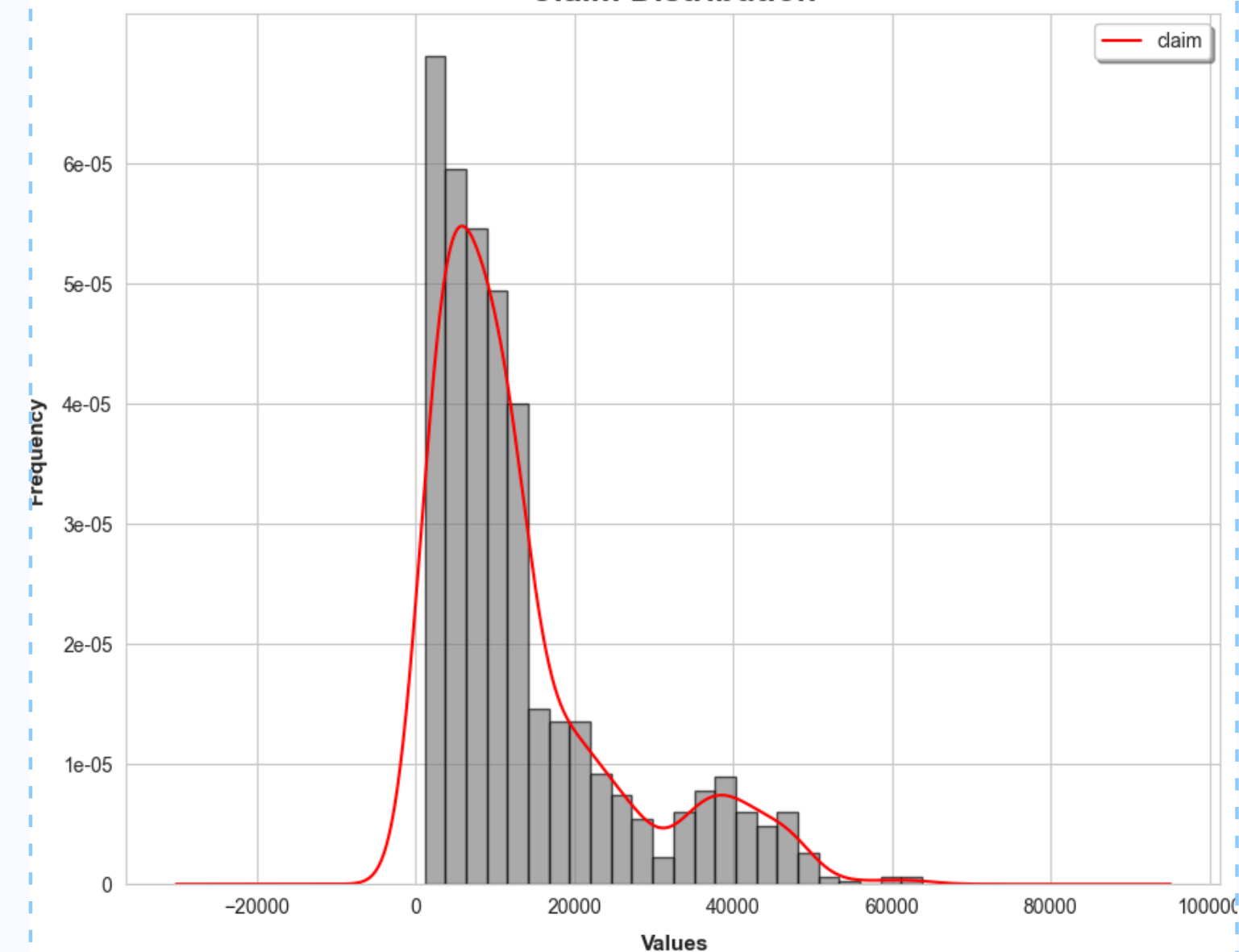
Claim: \$13,325

mean · median \$9,413

Histogram of Numerical Data



Claim Distribution



Claim is strongly right-skewed with bimodal structure → motivates tree-based ensemble models

EDA — Smoking Status Dominates

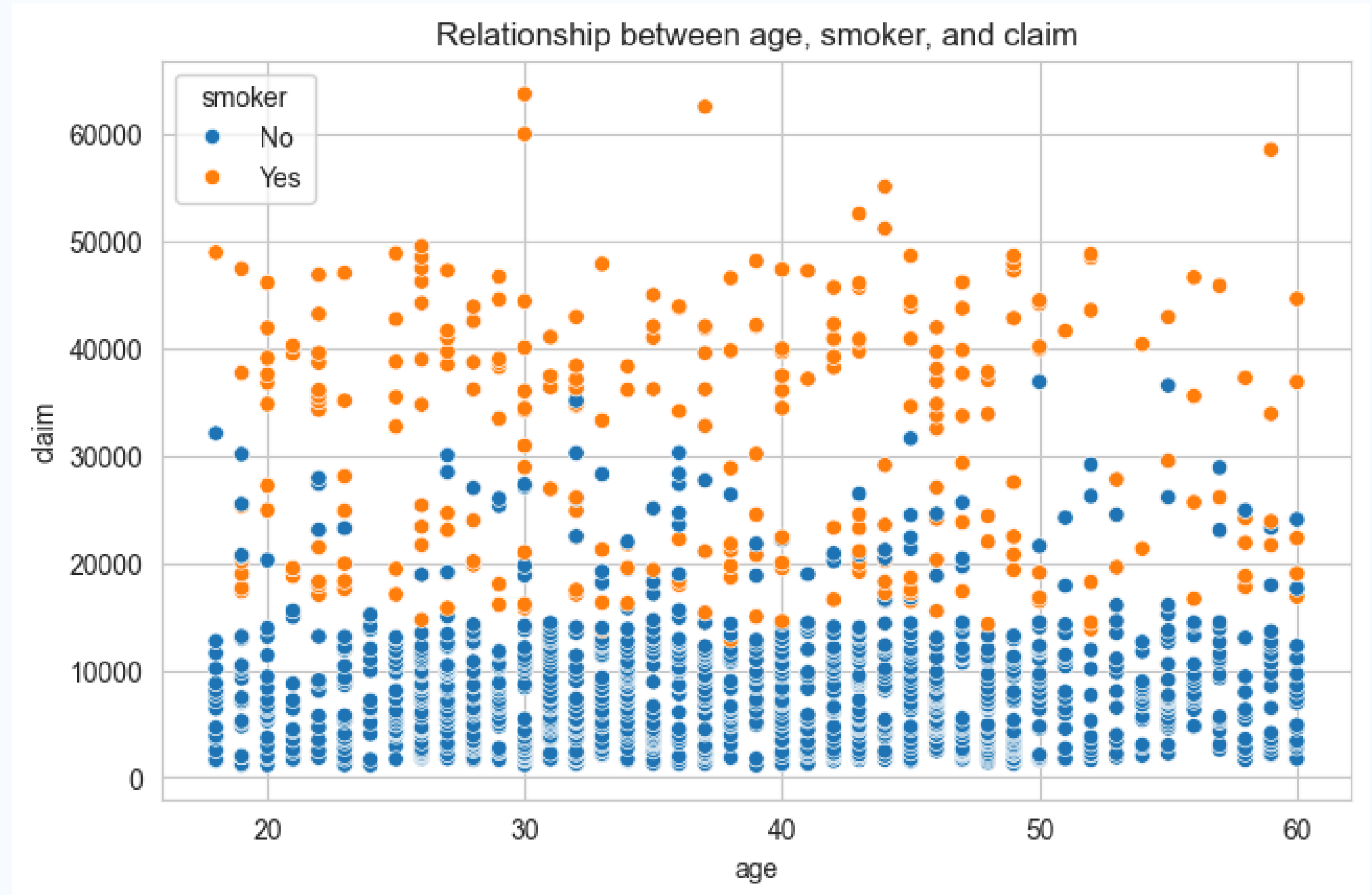
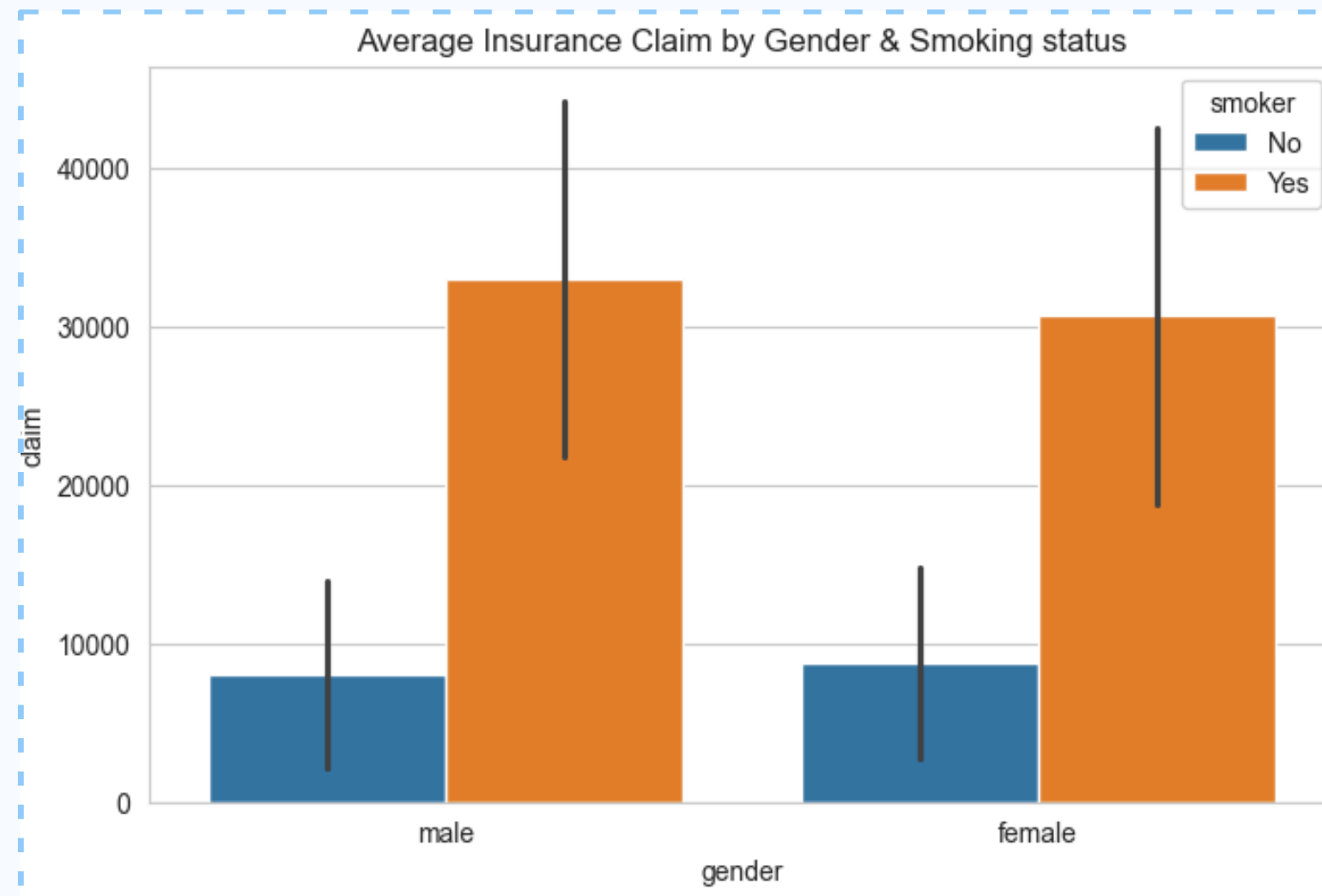
Smokers = 20.4% of population

but drive disproportionate share of total claims

Male smoker avg: **\$33,042**

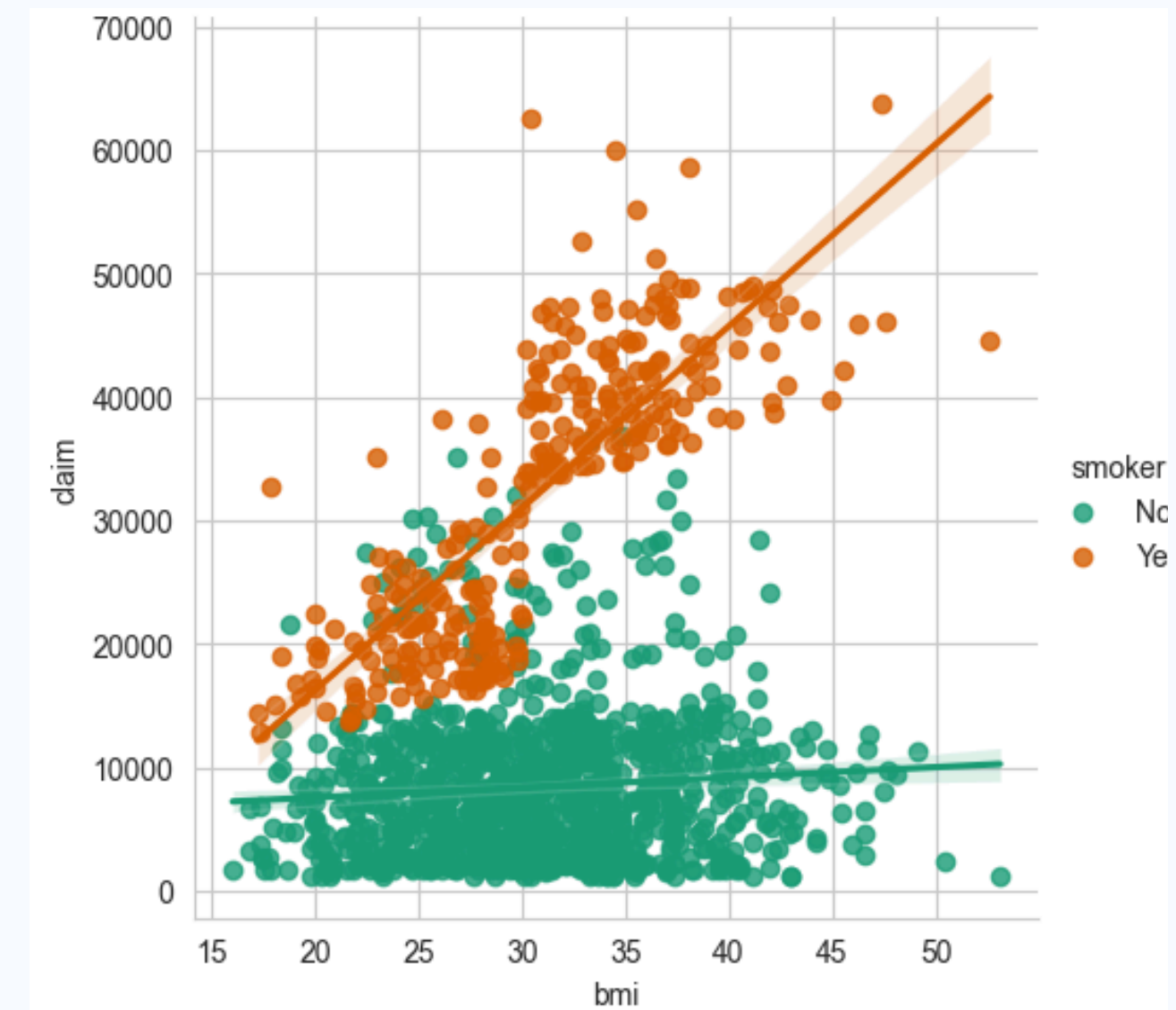
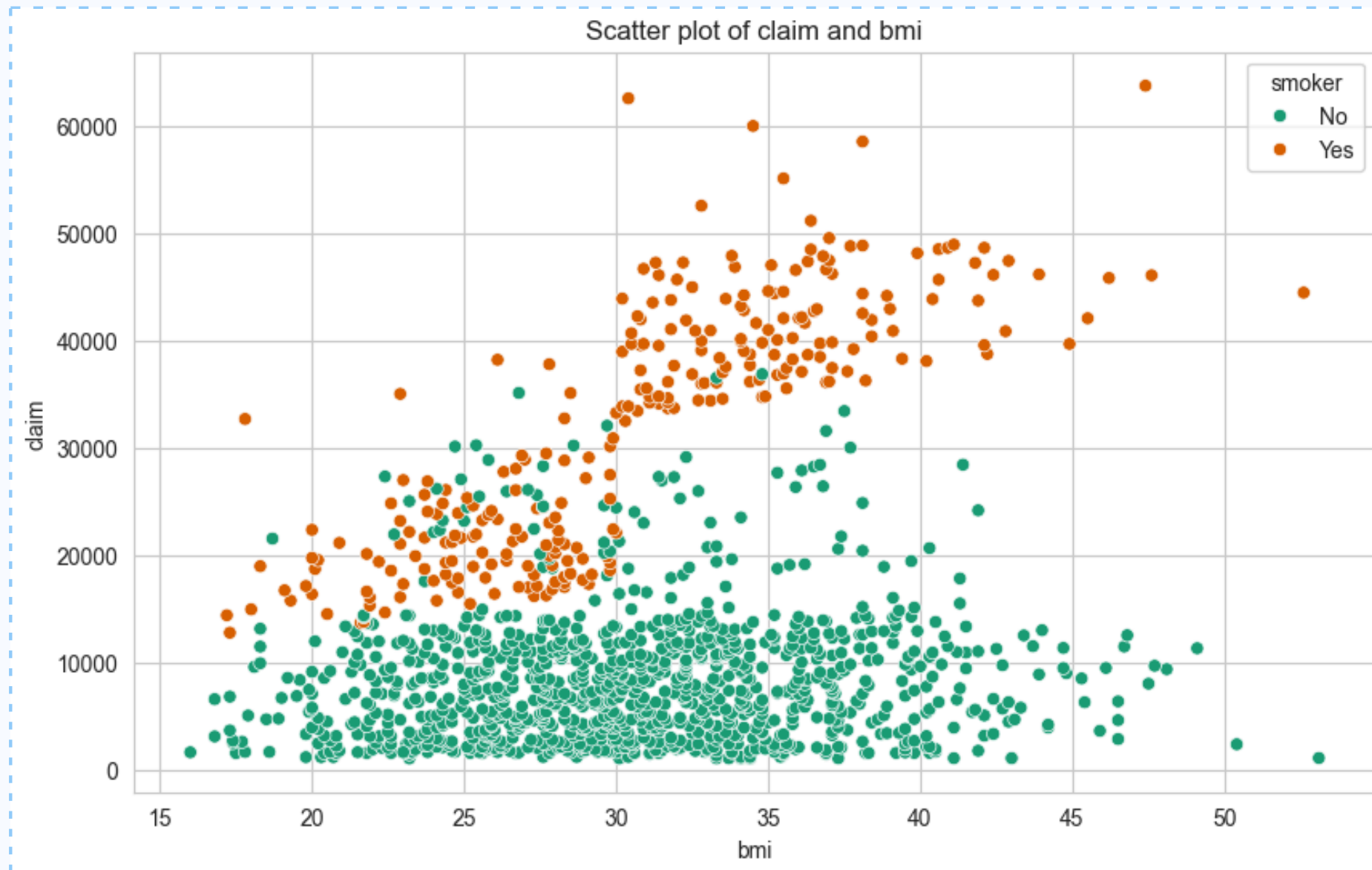
vs non-smoker:

\$8,062 (4.1× ratio)



Female smoker avg \$30,679 vs \$8,762 — 3.5× ratio. Gender differences within smoking groups are negligible.

EDA — The Critical Interaction: BMI × Smoking



Non-smokers

BMI slope: nearly FLAT

Smokers

BMI slope: STRONGLY POSITIVE

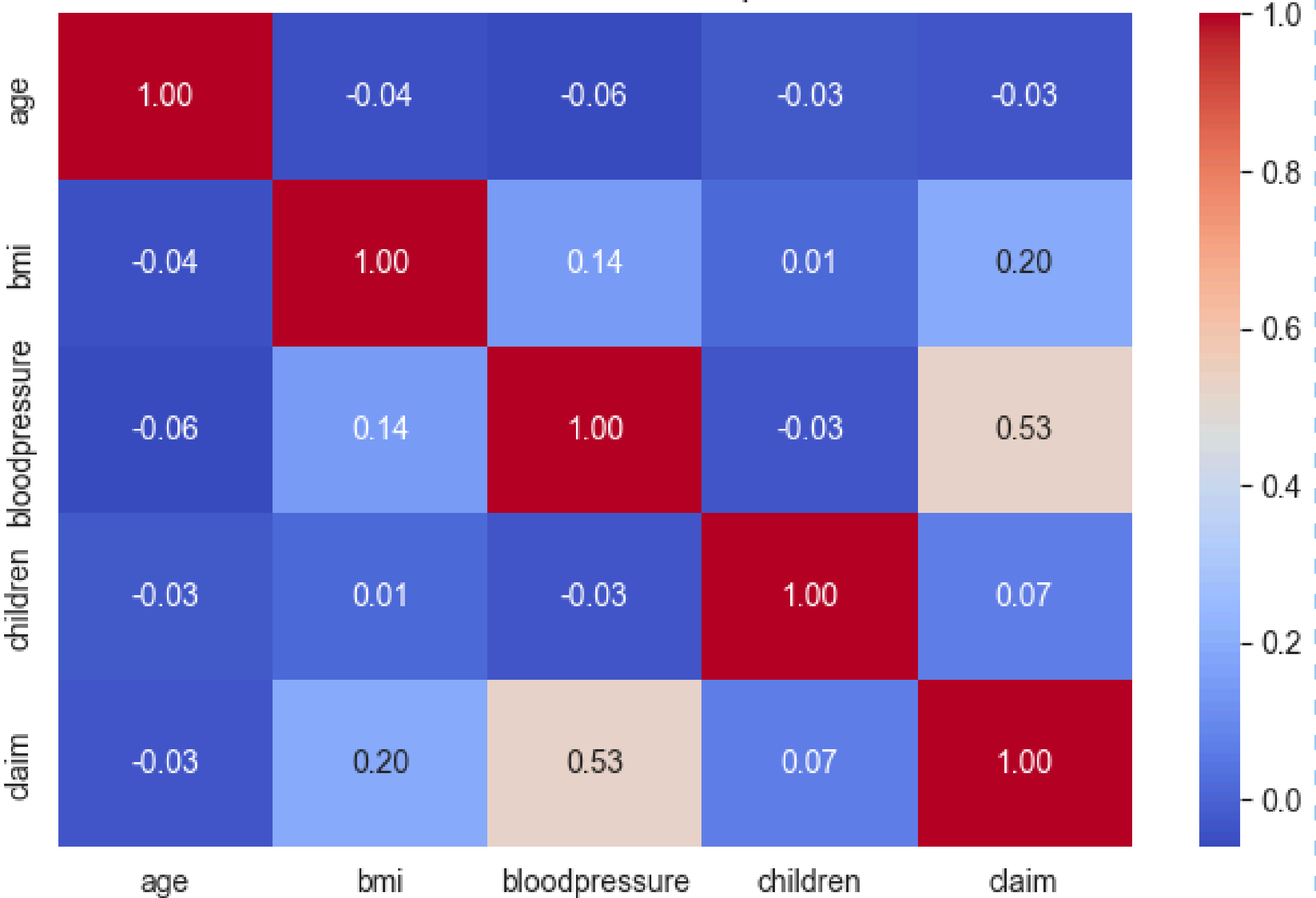
Why bmi_smoker matters

This motivates to create BMI_Smoker variable as $\text{BMI_smoker} = \text{BMI} \times \text{Smoker}$

EDA — Blood Pressure & Correlation Analysis



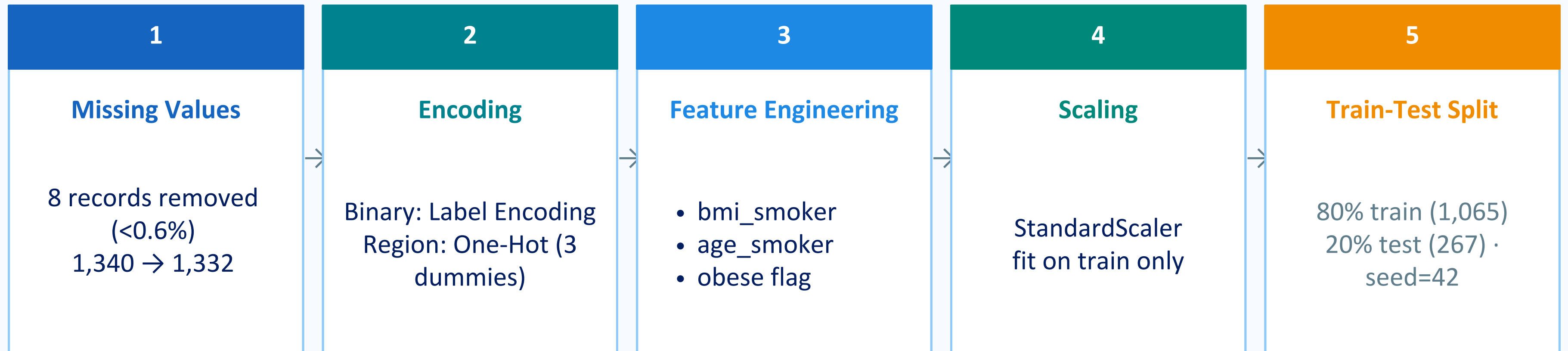
Correlation Heatmap



Correlation Summary

bloodpressure	$r = 0.53$
bmi	$r = 0.20$
age	$r = -0.03$
children	$r = 0.07$

Data Preprocessing & Feature Engineering



★ Feature Engineering

bmi_smoker

$= BMI \times smoker$

Captures the diverging slopes observed in EDA.

age_smoker

$= age \times smoker$

Captures whether the cost differential between smokers and non-smokers widens with age.

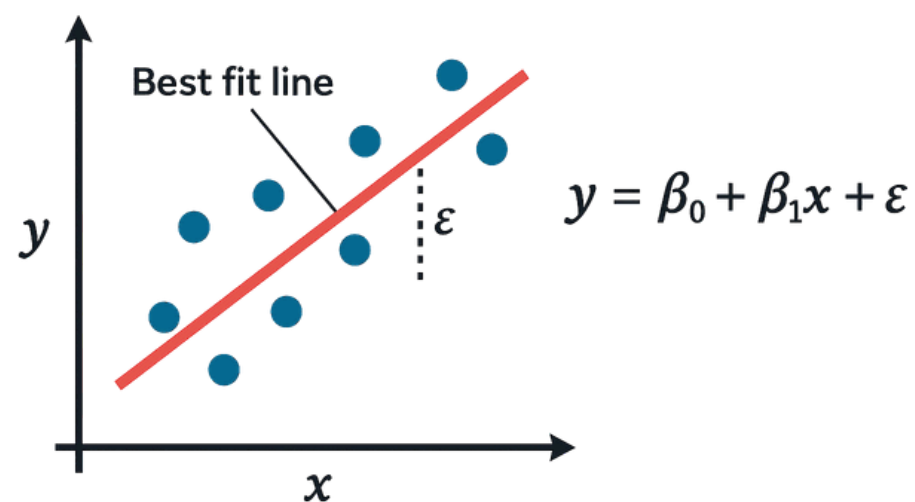
obese

$= 1 \text{ if } BMI \geq 30$

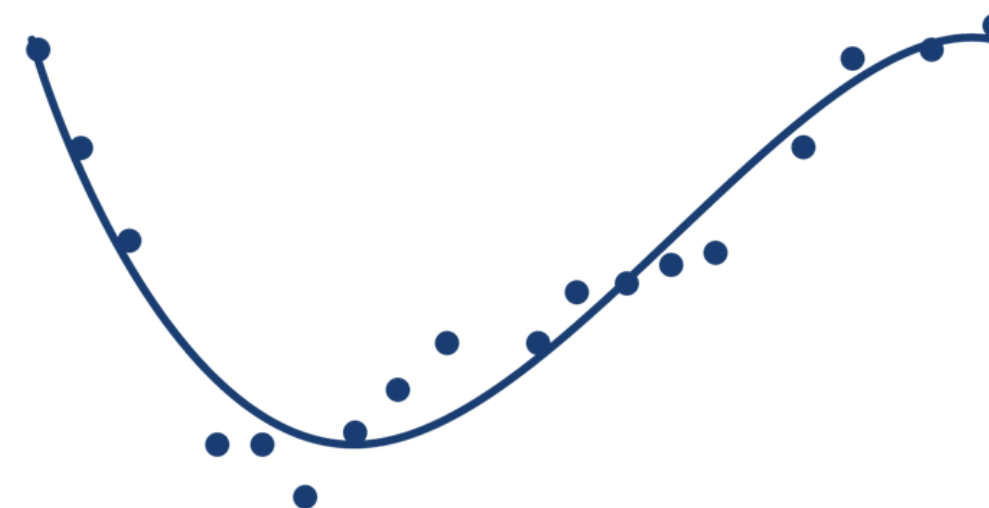
Explicit clinical threshold indicator consistent with WHO BMI classification guidelines.

Model Selection

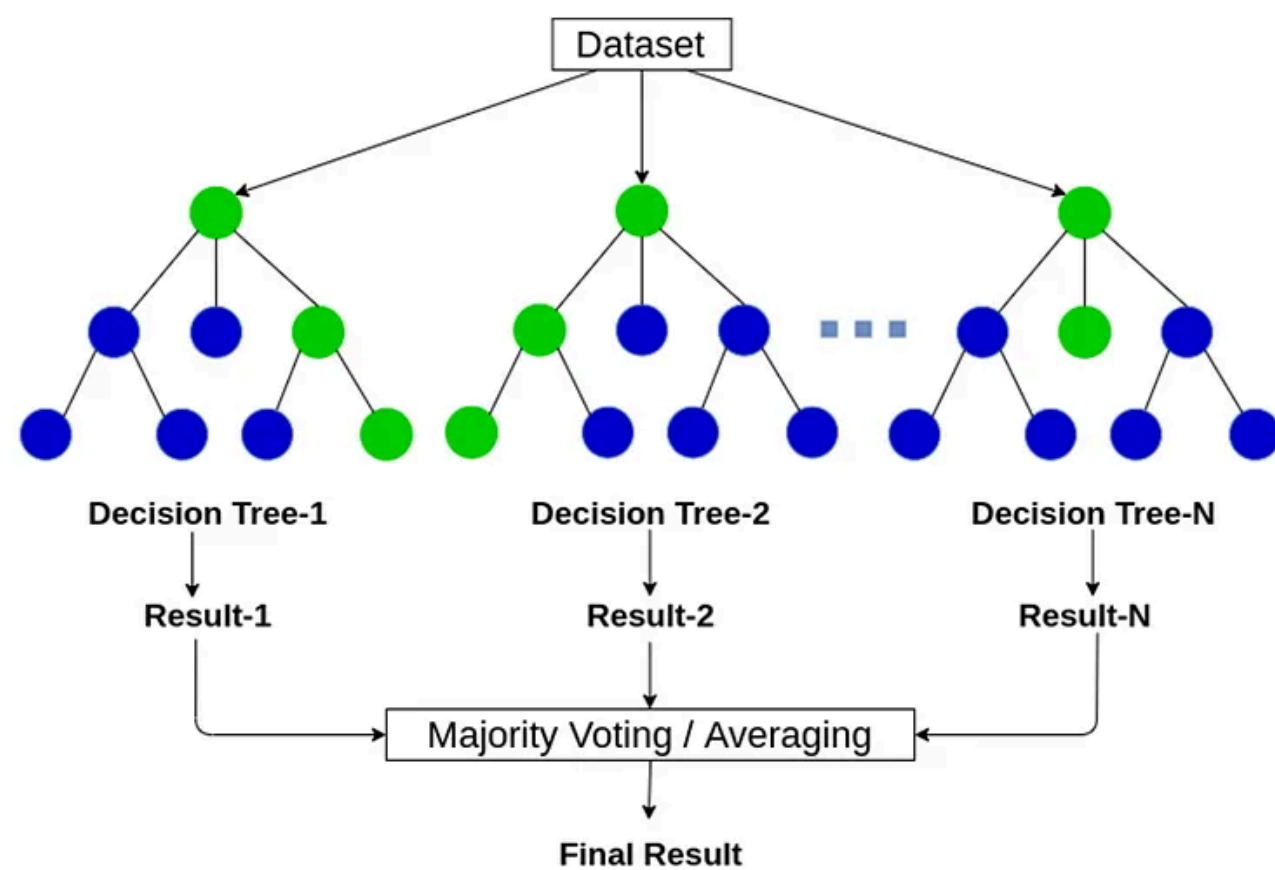
LINEAR REGRESSION



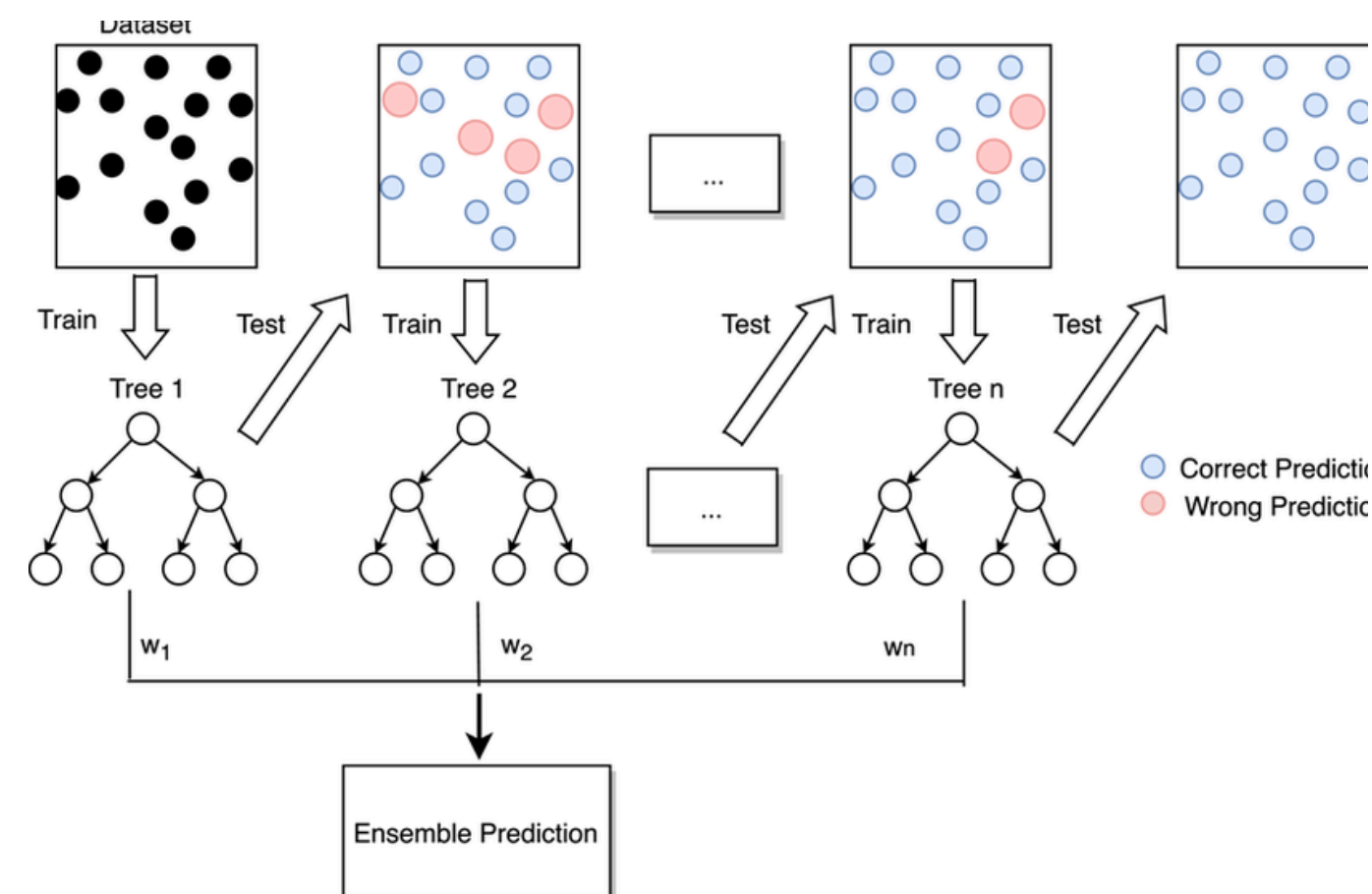
POLYNOMIAL REGRESSION



RANDOM FOREST



GRADIENT BOOSTING



Model Performance Results

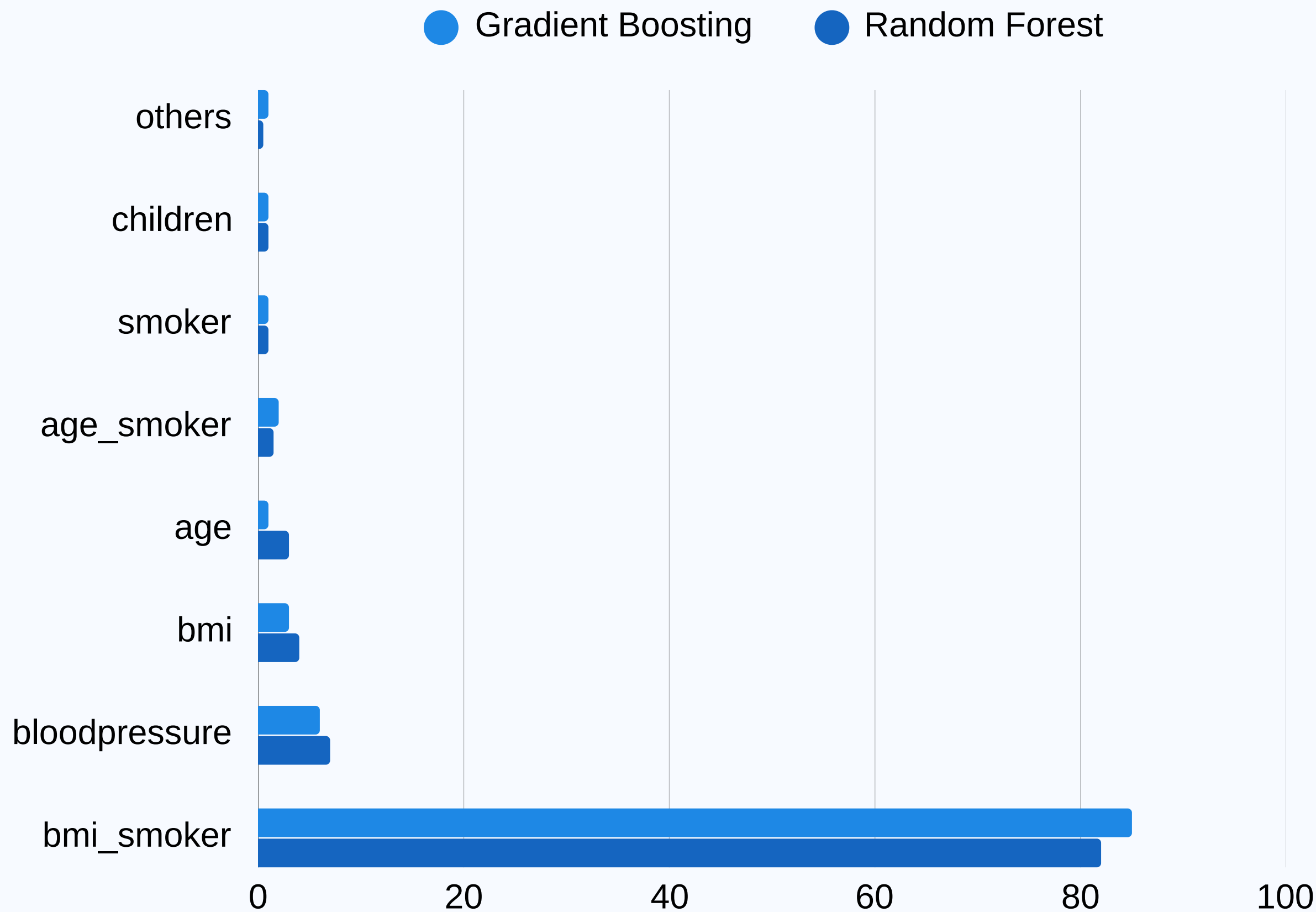
Model	R ²	MAE (USD)	RMSE (USD)
Linear Regression	0.797	\$4,185.51	\$5,408.97
Polynomial (deg=2)	0.811	\$3,949.72	\$5,226.38
Polynomial (deg=3) ⚠	0.311	\$5,827.77	\$9,965.82
Random Forest	0.831	\$3,787.74	\$4,939.55
Gradient Boosting ★	0.833	\$3,802.85	\$4,912.06

Summary

A test set of 267 observations was conducted.

- Gradient Boosting achieved the best performance, followed by Random Forest.
- Linear Regression achieved $R^2 = 0.797$ with feature engineering, up from a baseline of 0.672 without it — a gain of +0.125 attributable entirely to the bmi_smoker interaction term.
- Polynomial degree 2 improved modestly to $R^2 = 0.811$, while degree 3 collapsed to $R^2 = 0.311$ due to severe overfitting on a training set of 1,065 observations.

Feature Importance Analysis

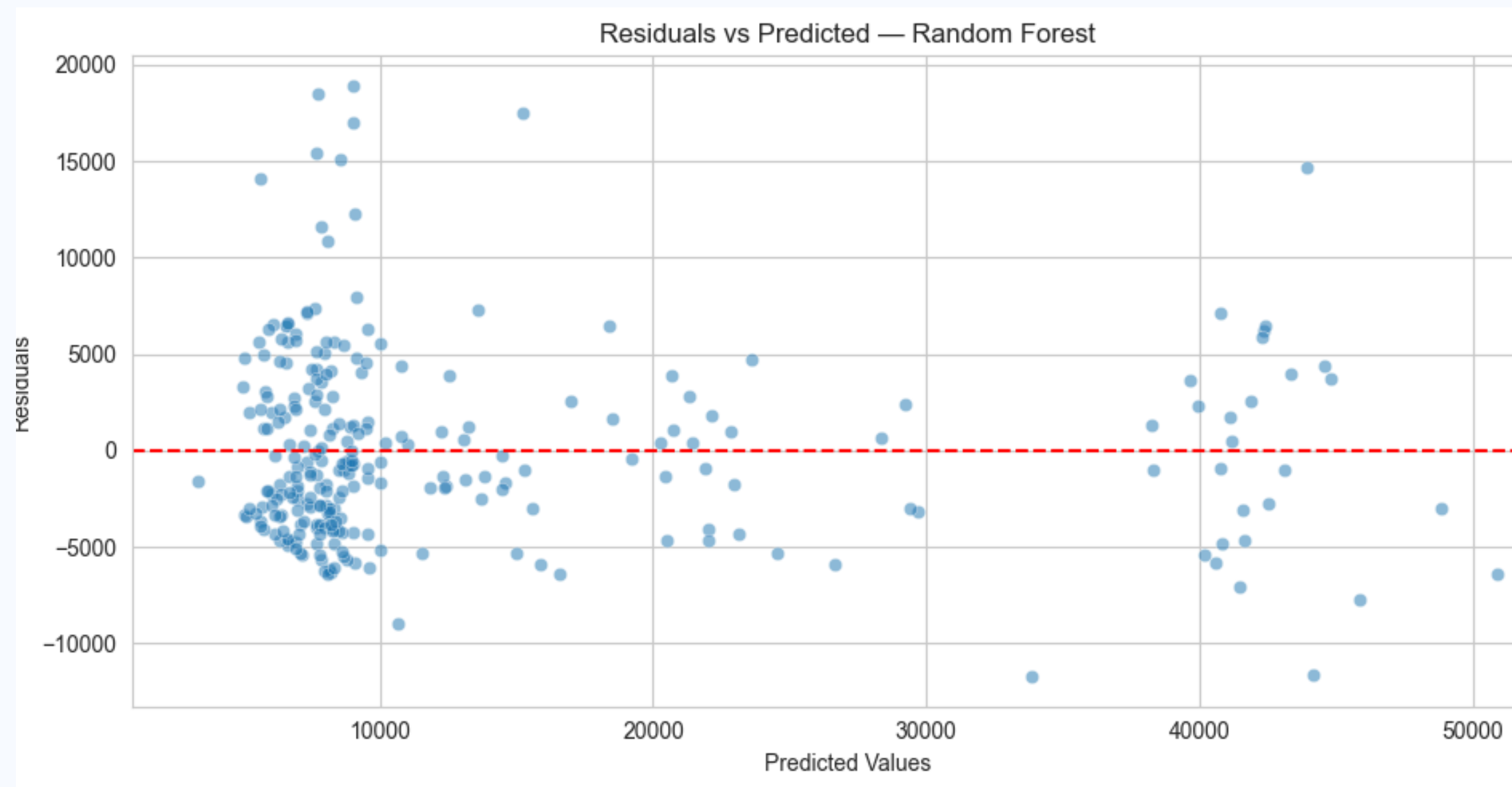


Key Findings	
82–85%	bmi_smoker Joint effect of BMI and smoking — dominant predictor in both RF and GB
6–7%	bloodpressure Independent cardiovascular risk signal
2–4%	bmi · age Residual effects after interaction absorbed
<1%	gender · region · diabetic · obese Negligible — after controlling for key risk factors
Both RF and GB converge on the same ranking — validates reliability of findings.	

Residual Analysis



A residual model was conducted, using the formula:
 $\text{Residual} = \text{Predicted cost} - \text{Actual cost}$



Findings

Low cost range (<\$15k)

High residual variance — model struggles with non-smokers incurring unexpectedly high costs (unobserved factors)

Mid range (\$15k–\$30k)

Residuals more evenly distributed around zero — model fits this band well

High cost range (>\$35k)

Residuals remain small — model captures the high-cost smoker-obese segment identified in EDA quite well

What-if Analysis — Practical Application

Base Profile:

Male · Age 45 · BMI 32 (obese) · Smoker · BP 90 · 2 children · Southeast ·
Non-diabetic

Predicted: \$42,317.64

Model	Predicted Cost	Change
Base case (smoker, BMI=32, age=45)	\$42,317.64	Baseline
Quit smoking	\$8,618.14	−79.6%
Reduce BMI to 25	\$21,107.89	−50.1%
Quit smoking + Reduce BMI to 25	\$8,583.17	−79.7%
Age 55 (same risk factors)	\$42,470.87	+0.4%
Age 55 + Quit smoking	\$8,998.47	−78.7%

−79.6%

Quit smoking

\$42,318 → \$8,618

−50.1%

Reduce BMI to 25

\$42,318 → \$21,108

−79.7%

Quit + BMI to 25

≈ quit alone

+0.4%

Age 55 (same factors)

\$42,318 → \$42,471



Statistical association — not a causal claim

Key Findings & Future Work

Key Findings

1. Gradient Boosting is the best model: $R^2 = 0.833$, RMSE = \$4,912
2. Feature engineering > algorithm: bmi_smoker gives +0.125 R^2 vs +0.036 R^2 from LR→GB
3. bmi_smoker accounts for 82–85% of feature importance in both ensemble models
4. Smoking cessation reduces predicted cost by 79.6%; BMI reduction alone gives 50.1%

Future Work

Vietnamese Data

Retrain on Vietnamese insurance claims once individual-level data becomes accessible

SHAP Analysis

Individual-level explainability — more rigorous than MDI importance

Clinical Variables

Add chronic disease history, medication records for higher predictive ceiling